Project Mnemos — Product Dossier

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Executive summary

Mnemos: Keepers of Veyra is now a working, local-first memory training RPG prototype with a complete 365-day curriculum, an adaptive engine that only counts delayed recall as mastery, and an original story whose progress is locked to demonstrated skill. It is ready for your review; nothing has been published.

- **Curriculum:** 12 modules, 52 weeks, 9 exercise templates, monthly gates and a day-1 baseline repeated every quarter (Section 4).
- **Learning engine:** every session opens with delayed recall; 10 documented adaptive rules; mastery requires recall after 1 and 7 days (Sections 5 and 6).
- **World:** Veyra's 12 districts are real memory-palace routes; Wren grows from apprentice to guide only when gates are passed (Section 7).
- **Prototype:** one offline HTML file, 28 unit tests and a full end-to-end test passing (Sections 12 and 13).
- **Independent audit:** 3 rounds, 29 defects found, 26 fixed, no release blockers; final overall score 7 of 10, engineering 8 (Section 12).
- **Decision needed:** approve the prototype for GitHub publication, or send changes. The largest remaining gap is story text for Chapters 4 to 12.

1. Research report

The strongest evidence says that 30 minutes a day of Method of Loci training can more than double word-list recall within six weeks, and that spaced retrieval is what keeps it. General "brain training" does not transfer, so Mnemos trains specific, named skills and never claims IQ gains.

Source note. Web search was disabled for this account during this session. Every source below is cited from established literature known before June 2026 and was not re-opened today. Figures are approximate and should be verified before public release.

1.1 What memory champions actually do

Source	Core method	What Mnemos takes from it
Dominic O'Brien, *How to Develop a Brilliant Memory Week by Week* (2005)	Journey method, Dominic System (digit pairs to person initials), 52-week structure	Journeys as the backbone; a person-based number system as an optional advanced track
Joshua Foer, *Moonwalking with Einstein* (2011)	Memory Palace, deliberate practice, the "OK plateau"	Plateau detection and deliberate difficulty increases
Nelson Dellis, *Remember It!* (2018)	Practical names, numbers, lists, PAO	Real-world transfer modules and the names protocol
Ed Cooke (co-founder of Memrise)	Playful, vivid, emotional imagery; "mems"	Tone: creativity over rote; user-generated images
Maguire et al., *Nature Neuroscience* (2003)	Superior memorizers had normal IQ and brain structure; most used Method of Loci	Evidence that the skill is learnable, not innate
Chase and Ericsson (1980s), subject "SF"	Digit span from about 7 to about 80 after roughly 230 hours	Skilled memory theory: chunking plus retrieval structures

1.2 Learning science that shapes the engine

Finding	Key source	Design consequence
Mnemonic training in novices: about 26 to 62 of 72 words after 40 days of 30 minutes; about 48 still recalled 4 months later	Dresler et al., *Neuron* (2017)	The 30-minute daily format is validated; Month 2 delivers the biggest jump
Method of Loci has large effects for ordered recall	Twomey and Kroneisen, *QJEP* (2021) meta-analysis	Palace is the backbone skill

Finding	Key source	Design consequence
Retrieval practice beats re-study	Roediger and Karpicke (2006)	Every session opens with recall, not re-reading
Optimal gap grows with the retention goal	Cepeda et al. (2006, 2008)	Probes at 1, 7, 30, 90 and 180 days
Interleaving improves discrimination and transfer	Rohrer and Taylor (2007)	Mixed review from Month 3 onward
Desirable difficulties	Bjork and Bjork (2011)	Difficulty rises only after stable accuracy
Interactive imagery beats separate images; bizarreness helps only in mixed lists	Bower (1970); McDaniel and Einstein (1986)	Teach "interaction first, oddness second"
Self-generated cues are remembered better	Slamecka and Graf (1978)	Users write their own images
Practice testing and spacing: high utility. Keyword mnemonic: low utility for durable learning	Dunlosky et al., *PSPI* (2013)	Keyword method always paired with spaced retrieval
Working-memory training does not transfer to general ability	Melby-Lervåg and Hulme (2013); Simons et al., *PSPI* (2016)	No IQ or clinical claims
Habits form over a median of about 66 days; one missed day does not break them	Lally et al. (2010)	Forgiving streaks, no punishment
Expected tangible rewards can undermine intrinsic motivation; informational feedback does not	Deci, Koestner and Ryan (1999)	Rewards describe progress, not bribes
Autonomy, competence, relatedness drive lasting motivation	Ryan and Deci (2000)	Choice of path, visible mastery, a companion who learns too
Teaching a learner agent improves the teacher	Chase et al. (2009); Fiorella and Mayer (2013)	"Teach Wren" reflection step

1.3 Technique evaluation by the curriculum committee

Technique	Evidence	Everyday transfer	Difficulty	Verdict
Visualization and interactive imagery	Strong	High	Low	Foundation, Month 1
Link and story method	Moderate	Medium	Low	Foundation, Month 1
Method of Loci (Memory Palace)	Strong	High	Medium	Backbone, Month 2 onward
Name and face association	Moderate	Very high	Medium	Core, Month 3
Major System (phonetic numbers)	Moderate	High	High	Core, Months 4 and 5
Peg systems (number-shape, rhyme)	Moderate	Medium	Low	Supporting, Month 6
Keyword method (vocabulary)	Strong short term, weak long term alone	High	Medium	Core, only with spaced retrieval
Dominic System	Practitioner evidence	Medium	High	Optional advanced track
PAO (person-action-object)	Practitioner evidence	Low	Very high	Optional, competition track, Month 9
Playing-card memorization	Practitioner evidence	Very low	Very high	Excluded from core, optional challenge
Working-memory n-back drills	Weak transfer	Very low	Low	Excluded

1.4 Learning apps reviewed (principles only, no assets copied)

App	Principle that works	Mnemos version
Duolingo	Tiny daily unit, clear path, streak habit	30-minute session with 10-minute recovery mode; forgiving streak
Brilliant	Learn by doing, explanation after attempt	Try first, then explain
Elevate	Clean per-skill scores	Skill radar from real data
Khan Academy	Mastery levels per skill	Five-state mastery model with delayed checks
Habitica	RPG identity around habits	World restoration tied to mastery, not clicks

2. Committee positions and resolved disagreements

Each expert drafted a position on its own before seeing the others. Seven real conflicts came up. Six were resolved and one stays open for your decision.

2.1 Independent positions

Expert	Position in one line	Deliverable in this dossier
1. Curriculum Committee	Palace first, numbers second, competition formats last and optional	Section 4 curriculum
2. Learning Scientist	Retrieval before instruction; mastery only counts after a delayed check	Sections 5 and 6
3. Creative Director	The world itself is a memory palace; the companion learns alongside the user	Sections 3, 7, 8
4. UI/UX Designer	One primary action per screen; zero motion during encoding	Section 9
5. Frontend Engineer	React + Vite static build, IndexedDB with fallback, single-file prototype	Section 10

Expert	Position in one line	Deliverable in this dossier
6. Auditor	Not involved in design; tested the built prototype cold	Section 12

2.2 Disagreements and how they were settled

#	Conflict	Side A	Side B	Resolution
1	Numbers system	Committee: Dominic System, champion-proven	Learning Scientist: Major System, makes real words and transfers to dates and phone numbers	Major is core. Dominic and PAO become an optional advanced track in Month 9
2	Session order	Creative: open with story	Learning Scientist: open with retrieval of yesterday's material	Retrieval first. A two-line story beat frames it, under 20 seconds
3	Streaks	Creative: daily streak drives return	Learning Scientist and UX: streak loss punishes; Lally (2010) shows one miss does not matter	Streak kept, with one automatic rest day per week; headline metric is 14-day consistency
4	Story gating	Creative: unlock by days to keep pacing	Learning Scientist: unlock by mastery	Calendar advances lessons. Story chapters and companion growth unlock only by passing gate assessments
5	Adaptivity range	Engineer: full adaptive reordering	Committee: fixed syllabus order	Module order fixed. Adaptation changes difficulty, review mix and remediation, capped at 40% of a session

#	Conflict	Side A	Side B	Resolution
6	Speed scoring	Committee: speed is central in competition	Learning Scientist: speed before accuracy builds sloppy encoding	Speed weighted 10%, only timed after accuracy reaches 80%
7	Face images	UX: photo-realistic faces	Engineer and IP: licensing and privacy risk	Original procedural illustrated faces in v1. **Open:** licensed photo set in v2?

2.3 Cut as unnecessary complexity

- Accounts, logins and cloud sync in v1 (local-first; export file transfers devices).
- Leaderboards (social comparison hurts beginners; may return as opt-in).
- 3D game engine (cost and battery; CSS depth and SVG give the world its depth).
- Playing-card drills in core (low transfer).

3. Product strategy and creative direction

The product is **Mnemos: Keepers of Veyra**, a free, open-source, local-first memory training RPG for adults. The chosen concept scored 37 of 40 because the world is itself a memory palace: every district the user restores is a set of loci they actually use.

3.1 Strategy

Item	Decision
Target user	Adult beginner, no prior technique knowledge, 30 minutes a day
Promise	Measurable gains in specific memory skills, tracked honestly; never IQ or medical claims
Distribution	Free, open source, static web app (GitHub Pages after your approval)
Business model	None; donations optional later

Item	Decision
North-star metric	Delayed (7-day) recall accuracy across trained skills
Guardrail metrics	Session length stays at or under 30 minutes; no guilt messaging; no data leaves the device by default
Non-goals v1	Accounts, social feeds, leaderboards, ads, clinical assessment

3.2 Concept evaluation (1 to 5 per criterion)

Concept	Emotion	Narrative length	Fit with exercises	Originality	Feasibility	Adult tone	365-day support	Non-manipulative attachment	Total
E. Keepers of Veyra (Archive city + mystery hybrid)	5	5	5	4	4	5	5	4	**37**
A. Archive of Forgotten Worlds (brief)	4	4	4	3	4	4	4	4	31
C. Detective reconstructing forgotten events	4	3	4	4	3	5	3	4	30
B. Memory Academy	3	3	4	2	5	4	4	4	29
D. Sci-fi neural exploration	3	4	3	3	3	4	3	3	26

3.3 Why E won

- **Mechanic and story are the same thing.** Districts of Veyra are loci. Restoring a district means building and using a real palace route.
- **A mystery carries a year.** Archivist Calder vanished the night the Lethe Tide rose. Each mastery gate returns a fragment of what happened.
- **The ending teaches maintenance.** The final twist reframes forgetting as necessary; the climax asks the user to choose what to keep. That mirrors spaced retention after the course.

- **Adult tone.** Literary, melancholy and wry rather than cute. No mascots that cheer every tap.

The name "Mnemos" and the river Lethe come from Greek myth, which is public domain. Veyra, Wren, Calder and all story text are original.

4. The 12-month curriculum

The year runs as 12 modules, 52 weeks and 365 training days, built from 9 reusable exercise templates with data-driven parameters. The curriculum advances by completed training days, not calendar days, so a missed day never skips content.

4.1 Annual objectives

1. Ordered recall of 30 or more unrelated items via a palace, with at least 85% accuracy.
2. 7-day retention of at least 70% on every trained skill.
3. Names for 10 new faces at 70% or better.
4. 40 digits in 5 minutes at 85% or better via the Major System.
5. 20 vocabulary items per week retained at 80% after 7 days.
6. A personal maintenance plan in active use at day 365.

These are targets for a typical adult, informed by Dresler et al. (2017). They are not guarantees. Results are always reported against the user's own day-1 baseline.

4.2 Module matrix, part A: what is taught

Module	Weeks	Skill	Objective	Prerequisite	Technique	Lesson	Exercise	Difficulty
M1 Lamplighters' Quay	1 to 4	Imagery	15 words in order after 2 min, 80%	None; day-1 baseline	Attention, interactive imagery, link, story chain	Images that interact stick	Word Recall, Linked Pairs, Story Chain	1 to 3
M2 The Long Gallery	5 to 8	Palace	20 items forward and reverse at 90%; 7-day 70%	M1 Practiced	Method of Loci: route, loci, placement, reverse	A palace is a fixed route; recall is a walk	Palace builder, Palace Walk	2 to 5
M3 Hall of Faces	9 to 13	Faces	10 names in 2 min at 70%; 1-day 60%	M2 Practiced	Feature selection, name-to-image, spaced rehearsal	Feature, image, interaction	Names and Faces	1 to 5
M4 Counting House	14 to 17	Numbers	00 to 99 at under 3 s each, 90%	M1 Practiced	Major System	Digits become consonant sounds	Major Conversion, Digit Recall	1 to 6

Module	Weeks	Skill	Objective	Prerequisite	Technique	Lesson	Exercise	Difficulty
M5 Clockwork Market	18 to 21	Numbers	40 digits in 5 min at 85%	M4 Proficient	Number palace, dates, phone numbers	Two digits per image, two images per stop	Digit Recall	3 to 7
M6 Tongues Bridge	22 to 26	Vocab	20 words, 80% at 7 days	M1, M2 Practiced	Keyword, abstract substitution, pegs	Keywords need spaced retrieval	Vocabulary, abstract Linked Pairs	2 to 6
M7 Scriptorium	27 to 30	Structure	10 facts or a 7-point outline, 80% at 1 day	M2 Proficient	Outline to palace, cue-answer, self-explanation	Reduce, place, explain	Fact Recall, Story Chain	2 to 6
M8 Tidal Stair	31 to 34	Palace (speed)	30 words in 3 min at 85%	Timed accuracy 80%	Faster encoding, 50-stop route	Accept the first good image	Timed Palace Walk, timed Digits	4 to 8
M9 Mirror Wards	35 to 39	Palace (robustness)	80% after a competing list	M8 Practiced	Palace reuse, separate palaces, mixed recall, optional PAO	Interference is the enemy of reuse	Interference Word Recall, mixed review	4 to 8
M10 Open Roads	40 to 43	Structure (transfer)	75% on unfamiliar formats	Q3 done	Technique selection, book and exam study	Choosing is a skill	Transfer tasks	3 to 7
M11 Amphitheatre	44 to 47	All	5 events each at 75%	All Practiced	Pacing, recall under pressure	Keep the routine	Mini-decathlon	5 to 9
M12 Lethe Gate	48 to 52	All	Annual battery and maintenance plan	M11 attempted	Retention maintenance, curation	Forgetting is the filter	Annual battery	Mixed

Every module runs in the 30-minute daily format (Section 5).

4.3 Module matrix, part B: how it is measured and what it unlocks

Module	Assessment	Mastery threshold	Retention test	Remediation	Transfer	Narrative	Unlock
M1	15-word chain plus 10 pairs	Gate 80%; 1-day 70%	1, 7, 30 days	Imagery coaching, smaller lists, written images	Errand list	Prologue, Ch. 1: relight the quay	Long Gallery
M2	20 items forward then reverse	90% fwd, 85% rev; 7-day 70%	1, 7, 30, 90 days	Route review, 5 loci, locus check	Tomorrow's to-do list	Ch. 2: Calder walked this route nightly	Hall of Faces
M3	Gate plus Q1 battery	Gate 70%; Q1 word recall 1.5x baseline	1, 7 days	Feature-only, then name-image drills	3 names at your next meeting	Ch. 3: one face missing from every wall	Counting House; Q1 report
M4	20 random pairs, timed	90% under 3 s; 7-day 85%	Expanding to 180 days	Decade blocks; rewrite weak images	A code you already know	Ch. 4: ledgers stop mid-sum	Clockwork Market

Module	Assessment	Mastery threshold	Retention test	Remediation	Transfer	Narrative	Unlock
M5	40 digits, 5 min	85%; 7-day 70%	1, 7, 30 days	Back to conversion; 20 digits	Work phone numbers	Ch. 5: the Tide runs on a schedule	Tongues Bridge
M6	Gate plus Q2; 6-month palace probe	7-day 80%	1, 3, 7, 14, 30, 60 days	User rewrites keywords	Your own vocabulary	Ch. 6 twist: Calder opened the gate	Scriptorium; Q2 report
M7	12 facts plus outline	80% at 1 day	1, 7, 30 days	Fewer facts; make-your-own-question	5-min talk from memory	Ch. 7: the copies disagree	Tidal Stair
M8	30 words, 3 min	85% at target time	1, 7 days	Study time restored if accuracy under 75%	Agenda in 2 min	Ch. 8: race the tide	Mirror Wards; PAO
M9	Gate plus Q3	80% under interference	All gates at 90 days	Fresh palace; 2-day rest	Two clients kept apart	Ch. 9: false memories	Open Roads; Q3 report
M10	Unfamiliar mixed task	Transfer 75%	7, 30 days	Technique-selection coaching	Is the module	Ch. 10: memories from outside	Amphitheatre
M11	Decathlon	Each event 75%	1, 7 days	Targeted week on weakest event	Exams, interviews	Ch. 11: the Echo of Calder	Lethe Gate
M12	Annual vs day-1 baseline	Reported, not pass or fail	Weekly maintenance	Plan targets weakest skill	Maintenance plan	Ch. 12: what Veyra keeps	Keeper mode

4.4 The 52-week plan

Drill notation: template, item count, study seconds. Counts are shown at difficulty 1; the adaptive engine raises them up to 60% as the user improves. Day 7 of every week is the weekly check; the last day of each module is its gate.

Wk	Mod	Focus	Objective	Practice drills	Day-7 check
1	M1	Attention and vivid images	Specific image per word in under 6 s	Word Recall 8 / 60s	Word Recall 8 / 50s
2	M1	Linking pairs	One cue brings back the other	Linked Pairs 8 / 70s	Linked Pairs 8 / 60s
3	M1	Story chains	12 items in order	Story Chain 10 / 90s	Story Chain 10 / 80s
4	M1	Consolidation and gate	Choose imagery vs chain	Chain 12 / 90s; Pairs 10 / 70s	Word Recall 12 / 70s

Wk	Mod	Focus	Objective	Practice drills	Day-7 check
5	M2	Building your first palace	10-stop route	Palace 5 / 60s	Palace 5 / 50s
6	M2	Placing and walking	10 items by walking	Palace 10 / 90s	Palace 8 / 70s
7	M2	Reverse and random access	Backward recall	Palace 12 / 100s reverse	Palace 10 / 80s
8	M2	Expanding the gallery	20 items both ways	Palace 15 / 120s; Palace 10 / 80s reverse	Palace 15 / 110s
9	M3	Seeing a face	Stable feature in 3 s	Faces 4 / 50s	Faces 4 / 45s
10	M3	Names as images	Names become pictures	Faces 6 / 60s	Faces 5 / 55s
11	M3	Faces at volume	8 names in 2 min	Faces 8 / 90s	Faces 6 / 70s
12	M3	Mixed review	Interleave skills	Faces 8 / 80s; Palace 15 / 100s	Chain 12 / 80s
13	M3	Quarter 1 assessment	Growth vs baseline	Word Recall 15 / 90s; Faces 8 / 80s	Palace 20 / 150s
14	M4	Major System 0 to 9	Ten sound codes	Major 10	Major 10
15	M4	Major 00 to 49	Fixed images	Major 15	Major 12
16	M4	Major 50 to 99	Fixed images	Major 15	Major 15
17	M4	Conversion speed	Under 3 s per pair	Major 20 timed; Digits 8 / 60s	Major 20 timed
18	M5	Digits in the palace	16 digits	Digits 12 / 90s	Digits 12 / 80s
19	M5	Longer numbers	24 digits in 3 min	Digits 20 / 150s	Digits 16 / 120s
20	M5	Dates and phone numbers	Real formats	Digits 10 / 70s; Pairs 8 / 70s	Digits 10 / 60s
21	M5	Number gate	40 digits in 5 min	Digits 24 / 180s	Digits 24 / 160s
22	M6	Keyword method	Sound-alike links	Vocab 6 / 90s	Vocab 6 / 80s
23	M6	Abstract to concrete	Substitutes	Pairs 8 / 80s abstract; Vocab 8 / 90s	Pairs 8 / 70s abstract
24	M6	Peg systems	Number-shape pegs	Chain 10 / 80s; Vocab 10 / 100s	Vocab 10 / 90s
25	M6	Vocabulary volume	12 words at 7 days	Vocab 12 / 120s	Vocab 10 / 100s
26	M6	Quarter 2 assessment	Half-year check	Word Recall 15 / 90s; Digits 16 / 120s	Vocab 10 / 90s

Wk	Mod	Focus	Objective	Practice drills	Day-7 check
27	M7	Facts as cue and answer	Capitals, element symbols	Facts 6 / 90s	Facts 6 / 80s
28	M7	Outlines in the palace	7-point outline	Chain 7 / 70s; Facts 8 / 100s	Facts 8 / 90s
29	M7	Self-explanation	Explain in own words	Facts 10 / 110s	Facts 8 / 90s
30	M7	Structure gate	12 facts at 1 day	Facts 12 / 120s	Facts 12 / 110s
31	M8	Encoding faster	First usable image	Palace 15 / 90s timed	Palace 15 / 80s timed
32	M8	A 50-stop route	Extend palace	Palace 20 / 120s timed	Palace 18 / 100s timed
33	M8	Timed numbers	24 digits in 2 min	Digits 20 / 100s timed	Digits 20 / 100s timed
34	M8	Speed gate	30 words in 3 min	Palace 25 / 150s timed	Palace 25 / 150s timed
35	M9	Interference	List A after list B	Word Recall 10 / 70s interference	Word Recall 10 / 60s interference
36	M9	Separate palaces	Palaces by subject	Palace 15 / 100s; Word Recall 12 / 70s interference	Palace 15 / 90s
37	M9	Mixed category recall	Interleave all	Faces 6; Digits 12; Vocab 6	Facts 8 / 80s
38	M9	PAO (optional)	Person-action-object	Major 20 timed; Digits 18 / 100s	Digits 18 / 90s
39	M9	Quarter 3 assessment	Cumulative probes	Word Recall 15 interference; Faces 8	Palace 20 / 120s
40	M10	Choosing the technique	Right tool per material	Facts 8; Chain 10	Pairs 10 / 80s
41	M10	Learning from books	Chapter to outline	Facts 10 / 100s	Facts 10 / 90s
42	M10	Exam and technical material	Terms, procedures	Vocab 10; Chain 8	Vocab 10 / 90s
43	M10	Transfer gate	Unfamiliar tasks	Facts 10; Faces 8	Pairs 12 / 90s abstract
44	M11	Event pacing	Time plan per event	Palace 20 timed; Digits 20 timed	Faces 10 / 90s

Wk	Mod	Focus	Objective	Practice drills	Day-7 check
45	M11	Recall under pressure	Accuracy with a clock	Word Recall 20 timed; Vocab 10	Facts 12 / 100s
46	M11	Decathlon practice	All events	Palace 20; Faces 8; Digits 16	Vocab 10 / 80s
47	M11	Arena gate	Each event 75%	Palace 25; Faces 10	Digits 24 / 120s
48	M12	What to keep	Permanent set	Facts 10; Palace 15	Facts 10 / 90s
49	M12	Maintenance scheduling	Weekly 10-min review	Vocab 10; Digits 16	Faces 8 / 80s
50	M12	Annual battery I	Words, numbers, faces	Word Recall 15; Digits 24	Faces 10 / 90s
51	M12	Annual battery II	Vocab, facts, palace	Vocab 12; Facts 12	Palace 30 / 180s
52	M12	The Lethe Gate	Final assessment	Palace 30 / 180s; Faces 10	Digits 30 / 150s

Day 365 is the closing ceremony: annual report and the switch to Keeper mode.

5. Daily session and adaptive learning engine

Every session opens with retrieval of earlier material, because a delayed test is both the best practice and the only honest measure of retention. The engine adapts difficulty, review mix and remediation, but never the order of modules.

5.1 The 30-minute session (optimized)

Step	Minutes	What happens	Why
1. Recall warm-up	5	Recall sets from 1, 7, 30, 90 or 180 days ago with no re-study	Testing effect; measures real retention
2. Lesson	4	2 to 3 short cards; day 1 of a week teaches, other days refresh	Short instruction limits cognitive load
3. Guided practice	6	Smaller drill; user may type the image they formed	Scaffolding and generation effect
4. Independent practice	9	Full drill at the current difficulty	Deliberate practice at the edge of ability

Step	Minutes	What happens	Why
5. Check	4	New material in the week's check format	Near transfer; feeds the Transfer score
6. Reflect and teach Wren	2	Diagnose missed items; pick the right explanation for Wren	Metacognition; protege effect

This differs from the brief's draft (warm-up 3, assessment 4) in one way: retrieval moved to the start and doubled. Feedback is shown after each drill, not saved for the end, because immediate item-level feedback corrects errors before they consolidate.

5.2 Session types

Type	Length	When	Advances curriculum
Full	30 min	Default	Yes, one training day per calendar day at most
Recovery	10 to 12 min	Offered after 2 or more days away	No; counts for consistency
Review only	10 to 15 min	Any time; due probes plus mixed review	No
Assessment	20 to 30 min	Day 7 of each week; gate on the module's last day	Yes
Free practice	Any	Training Grounds, any unlocked drill	No; XP capped at 50 a day

5.3 Per-skill state

Each of the 7 skills keeps: difficulty level 1 to 10, a moving accuracy average (EWMA, alpha 0.35), session count, consecutive high and low counts, error counts by type, its delayed-probe history, and a mastery state.

State	Rule
0 Untrained	No attempts

State	Rule
1 Learning	Any attempt
2 Practiced	Average accuracy at least 70% over 3 or more sessions
3 Proficient	Average at least 85% and a 1-day-plus probe at least 80%
4 Mastered	Proficient, plus a 7-day-plus probe at least 80%, plus the module gate passed

Knowing a technique is never enough: states 3 and 4 require delayed recall. This is how the app separates explaining a palace from using one.

5.4 Adaptive rules

Rule	Trigger	Action	Rationale
R1 Level up	Accuracy at least 90% in 3 consecutive sessions at this level (timed drills: also within target time)	Difficulty +1: about 6% more items, slightly less study time	Desirable difficulty once performance is stable
R2 Level down	Accuracy below 50% in 2 consecutive sessions	Difficulty -1 (minimum 1)	Avoid cognitive overload
R3 Remediation	Average below 60% after 3 or more sessions	Next independent drill becomes a remediation drill: 40% fewer items, 50% more study time, coaching card	Rebuild encoding before load
R4 Error-pattern switch	Same error type 3 or more times across the last 2 sessions	Alternate explanation card and format (table below)	A repeated error means the explanation did not land

Rule	Trigger	Action	Rationale
R5 Spaced probes	Every practice set	Re-tested at 1, 7, 30, 90, 180 days. Pass (at least 80% of the immediate score, or 70% absolute) moves to the next interval; fail resets to 1 day with a re-learn pass	Gap grows with the retention goal (Cepeda et al.)
R6 Forgetting detection	Mean of the last 3 probes for a skill below 60%	Mastery drops one state; that skill's review frequency doubles for 7 days	Mastery must be maintained
R7 Less repetition	Skill at Mastered	Appears only in due probes and in mixed review at most once a week; easy drills earn 25% XP	Stop re-teaching what is known
R8 Interleaving	From week 9	One review slot per session draws from a different earlier skill	Discrimination and transfer
R9 Time cap	Always	Remediation plus review never exceed 40% of a session	Protects curriculum pace
R10 Gate	Module's last day	Pass: chapter, district restored, Wren grows. Fail: calendar continues, retry at most every 3 days, story waits	Story tracks demonstrated mastery

5.5 Error diagnosis

Signal	Likely cause	Response
Omission at immediate recall	Encoding: no image formed	"Write your image" scaffold; slower study
Wrong item in the right slot	Weak or ambiguous image	Distinctiveness card
Right items, wrong order	Sequence or route problem	Route rehearsal card
Correct untimed, fails timed	Speed	Return to untimed until 80%
Correct immediately, fails at 1 to 7 days	Retention	Earlier probe and a re-learn pass
User reports distraction	Attention	Short focus reset before study; no penalty

The user confirms the cause on the Reflect step. Self-report combined with the error type is used, never error type alone.

6. Assessment and scoring

The Memory Training Score (MTS, 0 to 1000) weights retention highest at 30%, because durable recall is the product's goal. Each component is shown on its own, and any component without enough data is left out and the weights re-normalized rather than guessed.

6.1 Assessment calendar

Level	When	Content
Daily check	Every full session	New material in the week's check format
Delayed probes	1, 7, 30, 90, 180 days	Earlier practice sets, no re-study
Weekly	Day 7	Mixed check of the week plus one earlier skill

Level	When	Content
Monthly gate	Last day of each module	Module mastery test, framed as the district's Echo
Quarterly	Weeks 13, 26, 39	Parallel form of the day-1 baseline battery
Annual	Weeks 50 to 52	Full battery vs baseline plus cumulative probes

6.2 Component scores (each 0 to 100)

Score	Definition	Window
Accuracy	Difficulty-weighted mean accuracy of practice and check drills	Last 14 days
Retention	Mean delayed-probe score relative to the immediate score, capped at 100%	Last 30 days
Speed	Target time divided by actual time on timed drills, only where accuracy is at least 80%	Last 14 days
Mastery	Mean mastery state across introduced skills (state 4 = 100)	Current
Transfer	Accuracy on check drills and unfamiliar formats	Last 14 days
Skill scores	Per-skill accuracy, retention and level	Per skill

6.3 Composite weighting

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\text{MTS} = 10 \times \frac{0.30R + 0.25A + 0.20T + 0.15M + 0.10S}{\sum w_{\text{available}}}
```

R is retention, A accuracy, T transfer, M mastery and S speed. Retention leads because it is the goal; accuracy before speed because speed without accuracy builds sloppy encoding; transfer is weighted above mastery because a technique that only works on app material has little value.

6.4 What the scores are not

The app states on the dashboard and in the About screen: scores measure performance on Mnemos tasks only. They are not an intelligence test, a clinical memory assessment or a diagnosis. Anyone concerned about memory changes should speak to a doctor.

7. Creative bible (v0.2)

Veyra is now a full world: a tiered city on a grey northern coast, nine centuries of history, a drug economy built on forgetting, and a year-long thriller in which the player turns out to be carrying the city's lost memory. About 24,000 words of original story, lore and lesson text ship in the app. The style takes its cue from epic fantasy: long, descriptive passages, landscapes and history, plain natural dialogue.

7.1 World and lore

Element	What it is
The Lethe and the Tide	A river whose evening mist takes whatever no one is holding in mind
The Lamp-Wardens	Founders who built Veyra itself as twelve memory-palace routes and swore the Compact of Lamps
The Long Quiet	Three centuries of decline: the city trusted paper ledgers, and the Art faded
Lethewater, "the grey"	Distilled river water that erases chosen memories; sold to the grieving and the poor
The Salt Guild	Merchants who run the grey trade behind a salt business
The Quiet Order	Gentle, grey-hooded preachers who teach that forgetting is mercy, funded by the Guild
The Night of No Moon	Once in 19 years the Tide becomes a wall; the old Wardens held the Gate through the Long Vigil
The Heartledger	The city's one true record, kept only in an Archivist's memory palace

The app holds 16 lore entries. They unlock through gates and side quests.

7.2 Hero, sidekick, villain

Role	Character	Arc
Hero	You, the Keeper	A stranger with no past discovers they are Calder's courier and carry the Heartledger; they chose this burden once and choose it again
Sidekick	Wren Halloway	A funny, anxious clerk who lives by notecards; learns she is Calder's hidden heir; ends as Archivist
Villain	Sevrin Ashgrave, the Grand Hush	A grieving father who drank away his loss and now wants to "free" the whole city; Calder's brother; redeemed at the Gate by a memory of his daughter
Mentor	Elsabet Calder	The vanished Archivist who opened the Gate on purpose; secretly living as Old Tansy, memory wiped
Drifter	Joss Hask	A huge, quiet road-warden hunting the men who hollowed out his brother
Intriguer	Isolde Varre	Harbour Mistress and card shark; apparent betrayer, actually Calder's spy in the Order
Doubter	Brother Ennick	A young acolyte who defects with the true record

Character profiles reveal more as gates are passed, so the twists stay hidden until earned.

7.3 Story structure

Ch.	District	Main quest	Turn
Prologue	The Archive	Find the Archive	The lantern lights for you alone
1	Lamplighters' Quay	Relight the lamps	The Hushed snuff them; first ledger page
2	Long Gallery	Walk Calder's route	Escape in the dark by walking it backwards

Ch.	District	Main quest	Turn
3	Hall of Faces	Restore the names	Hask arrives; the Grand Hush knows your name
4	Counting House	Crack the number code	Guild money funds the Order; a date: the Night of No Moon
5	Clockwork Market	Restart the clocks	Assassination attempt; Isolde saves you
6	Tongues Bridge	Read the inscriptions	**Twist:** Calder opened the Gate; the city turns on the Archive
7	Scriptorium	Find the true account	She did it to stop thieves; "find the courier"
8	Tidal Stair	Race the Tide	**Twist:** Isolde betrays you, then reveals she is Calder's spy
9	Mirror Wards	Escape with your memories	**Twist:** the scratched face is Wren's; she is the heir
10	Open Roads	Find out who you were	**Twist:** your own handwriting in a waystation register
11	Amphitheatre	Win the Rite of Echoes	**Twist:** Sevrin is Calder's brother; Old Tansy is Calder
12	Lethe Gate	Hold the Long Vigil	You walk every route as one road; Sevrin closes the Gate

There are three layers. The **major arc** is the 12 chapters, unlocked only by gates. The **minor arcs** are 52 weekly tales, one at the start of each week, which build suspense between gates. The **side arcs** are 24 side quests (Hask's brother, Isolde's games, Ennick's courage, Mosk's lost wife and more). Each finishes only when the player shows a real skill: a 7-day recall, a level reached, a perfect reverse walk. Each gate opens with a short boss-style encounter.

7.4 Wren's voice

Wren's lines were rewritten to sound like a real person: contractions, jokes, small admissions of fear. She never guilts the player. On a return after absence: "The lamps are still lit. I checked every night, which I'm slightly embarrassed to admit. A short session is a

perfectly good way back in."

7.5 Visual identity

Element	Decision
Palette	Ink night (#0F1420), lamplight amber (#E8B04B), sea-glass teal (#7FB3A6), parchment (#EFE6D2), Tide silver (#B8C2D6)
Type	Serif for story and headings, set large with a drop cap; system sans for UI; no external fonts
Illustration	Original flat SVG: lantern, isometric districts, fog. Restored districts gain light and colour
Motion	Lantern glow and fog drift outside drills only; none during study or recall

8. Gamification and rewards

Lumen (XP) is earned mainly by accurate recall at the right difficulty, and the single biggest payout is a successful delayed probe. Showing up alone earns little, and repeating easy drills earns a quarter of normal.

8.1 Lumen sources

Event	Lumen	Rule
Drill completed	10 x accuracy x (1 + 0.1 x (level - 1))	Accuracy-weighted, difficulty-scaled
Delayed probe	25 x probe score x interval factor (1 day 1.0, 7 days 1.4, 30+ days 1.8)	Retention pays most
Personal best on a skill	+20	Improvement, not volume
Skill level up (R1)	+30	Demonstrated progression
Full session completed	+15	Small consistency credit

Event	Lumen	Rule
Return after 3+ days, session done	+20 "Rekindled"	Rewards recovery, never punishes absence
Drill 2+ levels below the skill's level	25% of normal	No farming
Free practice	Capped at 50 a day	No grinding

8.2 Progression layers

Layer	Driven by	Visible as
Keeper rank	Total Lumen (rank = floor of the square root of Lumen/50, plus 1)	Rank number and title
Skill tree	Per-skill mastery states	Seven branches, nodes light by state
World map	Gates passed	Districts lose fog and gain colour
Wren	Gates passed	Six stages of her arc
Ledger fragments	Gates and mastery milestones	Calder's story, collectible
Achievements	Specific demonstrated feats	Seals in the Archive

8.3 Consistency without punishment

- The headline habit metric is **consistency: sessions in the last 14 days**, not an all-or-nothing streak.
- The streak still exists and is shown, with one automatic rest day per week that preserves it.
- A broken streak is described as "your longest run so far was N days"; no flames go out, no warnings.
- A recovery session (10 minutes) is offered after any absence.

8.4 Achievements in the prototype

First Light (first session) · Cornerstone (first palace built) · Clean Sweep (100% on a drill of 8+ items) · Walking Backward (perfect reverse palace walk) · It Stayed (1-day probe at 90%+) · A Week Later (7-day probe at 80%+) · Rekindled (return after 3+ days and finish) · Steady Flame (7 sessions within 14 days) · Honest Ledger (3 reflections with diagnosis) · Sharper (first skill level up) · Quay Restored (Month 1 gate) · Face Reader (8+ names correct in one drill) · Number Smith (10 of 10 Major conversions at under 3 s) · Safe Keeping (first profile export).

9. UI/UX specification

Mobile-first, five destinations, one primary action per screen. During study and recall the screen goes still: no animation, no companion chatter, no score counters, only the stimulus and a quiet timer.

9.1 Information architecture

Tab	Purpose	Key components
Today	Start the right session in one tap	Session card (Full, Recovery, Review), due probes count, Wren's line, week objective
Veyra	World and story	District map with fog, chapter list, ledger fragments
Palaces	Memory palace inventory	Palace builder, loci list, usage count, rest status
Progress	Honest analytics	MTS with components, skill radar, accuracy and retention trends, calendar heatmap, assessment history, weakest skills
Keeper	Profile and data	Avatar, rank, achievements, settings, export, import, delete, privacy, about

Mobile: bottom tab bar, 56 px targets. Desktop (900 px and up): left rail and a two-column Today screen.

9.2 Daily flow

1. Open app: Today shows one large "Begin day N" button and what is due.

2. Session runner: a thin step bar (6 segments) at the top; exit saves partial progress.
3. Each drill: Brief (what to do) → Study (stimulus only) → Recall (inputs) → Feedback (item-by-item, correct answers shown).
4. Reflect: tap a cause for missed items; choose the explanation Wren should hear.
5. Summary: scores, Lumen breakdown by source, any unlock, Wren's line, next due probe date.

9.3 Interaction patterns

Pattern	Spec
Typed recall	One field per slot; Enter moves to next; forgiving match (case, accents, plurals, one-letter typos on words of 5+ letters)
Ordered recall	Numbered slots; free recall uses a single list field
Palace recall	Each slot labelled with its locus name
Faces	Face shown, name typed below; illustrated faces only
Timer	Countdown shown as a thin bar plus seconds; "I'm ready" ends study early
Feedback	Green check / amber dash per item with the correct answer; never red flashes

9.4 Accessibility requirements

- WCAG 2.2 AA contrast in both themes; text at least 16 px; zoom to 200% without loss.
- Full keyboard path; visible focus rings; skip-to-content link.
- Screen reader: ARIA live region announces drill phase changes and results.
- prefers-reduced-motion removes all decorative motion; a manual toggle exists too.
- Colour is never the only signal: icons and text accompany every state.
- Timers can be extended 1.5x in settings for users who need more time; scores note the setting.

10. Technical architecture

React 18 with Vite, plain JavaScript, no backend. The build produces one self-contained HTML file for local review and a static folder for GitHub Pages later. All data stays in the browser in IndexedDB, with localStorage as fallback and a JSON file for backup and device transfer.

10.1 Framework decision

Option	Reliability	Maintainability	Pages compatible	Speed to build	Verdict
React + Vite	High	High; large contributor pool	Yes, static build	High	**Chosen**
Vanilla JS	High	Low at this size (40+ screens and states)	Yes	Medium	Rejected
TypeScript	High	Highest	Yes	Medium	Planned for v2 once data shapes settle
Game engine (Phaser, three.js)	Medium	Medium	Yes	Low	Rejected for v1: battery, bundle size, accessibility
SVG + CSS for world art	High	High	Yes	High	Chosen for map, faces, lantern

10.2 Code structure

Layer	Files	Responsibility
Data	curriculum, content pools, story	Modules, weeks, word pools, Major list, faces, story text
Engine	session builder, scoring, adaptive, scheduler, rewards	Pure functions, unit-tested, no UI
Storage	store	IndexedDB with fallback, export, import, schema migration
UI	screens and components	Today, Session runner, Drills, Veyra, Palaces, Progress, Keeper

10.3 Data model (profile record)

Field	Contents
meta	schema version, curriculum version, created, last opened
user	alias, avatar, preferences (theme, reduced motion, timer multiplier)
progress	curriculum day, training dates, minutes, streaks, Lumen, gates passed
skills	per skill: level, average, sessions, highs, lows, error counts, mastery
results	every drill result: date, day, skill, template, role, items, accuracy, timings, errors
sets	stored practice sets with probe schedule (next due, interval index, history)
palaces	name, ordered loci, last used
majorList	the user's personal 00 to 99 images
achievements, story	unlocked ids with dates; chapters seen

10.4 Persistence, privacy and offline

Topic	Behaviour
Save	After every drill and step, so closing mid-session loses at most one drill
Storage	IndexedDB; falls back to localStorage; asks the browser for persistent storage
Export	Downloads a .json file with schema version and a checksum
Import	Validates structure and version, previews the profile, then replaces only on confirmation

Topic	Behaviour
Device transfer	Export on device A, import on device B. No automatic sync
Cloud sync (later)	Supabase is the best fit (open source, row-level security, free tier). Needs accounts, so opt-in only and a v2 decision
Personal data	Only an alias. No email, phone, location or tracking. Nothing leaves the device
Clearing browser data	Deletes progress; the app says so on the Keeper screen and nudges a monthly export
Delete	Keeper → Delete all data, with confirmation
Offline	The single file needs no network at all. The hosted version adds a service worker and web manifest for install and offline caching
Needs internet	Nothing in v1, except the first load of the hosted version

11. Intellectual property register

Every visual, character and story element is original. External items are open-source code libraries under permissive licences, public-domain myth and techniques, and uncopyrightable facts. No assets from Duolingo or any other app are used.

Item	Source	Licence	Attribution	Commercial	Modify	Distribute	Restrictions
React, React DOM 18.3.1	Meta, npm	MIT	Keep licence text	Yes	Yes	Yes	None
Vite 5, plugin-react	npm	MIT	Build tool only	Yes	Yes	Yes	None
vite-plugin-singlefile	npm	MIT	Build tool only	Yes	Yes	Yes	None
Vitest	npm	MIT	Dev only	Yes	Yes	Yes	None
Playwright	Microsoft, npm	Apache 2.0	Dev only, not shipped	Yes	Yes	Yes	Notice file if redistributed
Fonts	System-installed only	n/a	None	Yes	n/a	Not bundled	None

Item	Source	Licence	Attribution	Commercial	Modify	Distribute	Restrictions
"Mnemos", Lethe, Mnemosyne	Greek mythology	Public domain	None	Yes	Yes	Yes	Check trademark search before public launch
Method of Loci, peg systems, link method	Classical and traditional techniques	Public domain methods	Credit in research notes	Yes	Yes	Yes	None
Major System	17th to 19th century (Winckelmann, Feinaigle, Paris)	Public domain method	None	Yes	Yes	Yes	None
Dominic System, PAO	Practitioner methods	Methods are not copyrightable	Name O'Brien as originator in docs	Yes	Yes	Yes	Described generically in-app; do not use "Dominic" as a brand
Veyra, Wren, Calder, all story text	Original	Project licence (proposed MIT for code, CC BY 4.0 for content)	Project credit	Yes	Yes	Yes	None
Illustrated faces, lantern, map art	Original procedural SVG	Same as above	None	Yes	Yes	Yes	None
Word pools, Major word list	Original selection of common English words	Same as above	None	Yes	Yes	Yes	None
Swahili vocabulary, capitals, element symbols	Facts	Not copyrightable	None	Yes	Yes	Yes	Verify spellings before launch

Not used on purpose: the acronym SMASHIN' SCOPE (associated with a commercial program), photographs of real people, any third-party character or mascot.

12. Testing report and independent review

After three review rounds the independent auditor found no release blockers and scored the prototype 7 of 10 overall. Engineering reached the 8/10 target. The other four categories finished at 6 or 7, and the reasons are listed below rather than argued away.

12.1 Automated testing

Suite	What it covers	Result
Unit (Vitest, 28 tests)	Curriculum integrity, answer matching, Major decoder, scoring, rules R1 to R6, probe schedule, gates, Lumen cap, streak rest day, MTS bounds, export/import validation, quarterly battery, discard rules	28 of 28 pass
End-to-end (Playwright, 390 px phone)	Onboarding, day 1 baseline, reload persistence, day 2 with delayed recall, palace, faces and Major weeks, all tabs, export, delete, import, desktop layout, horizontal overflow, console errors	Pass, zero errors
Auditor scripts	Edge cases: leave and resume, double clicks, timeouts, interference, gate fail and retry, tampered imports, two open tabs, keyboard path	Pass in round 3

12.2 Review scores by round (1 to 10, target 8)

Category	Round 1	Round 2	Round 3
Curriculum and learning	5	6	7
Creative and gamification	6	6	6
UI/UX and accessibility	7	7	7
Technical engineering	5	7	8
Overall product	5	6	7

12.3 Defects found and fixed

Severity	Found	Fixed	Examples
Critical	1	1	A malformed backup could brick the app; now fully validated, plus an error screen with a raw-data download
High	5	5	Probes could be retaken after seeing answers; two tabs overwrote each other; levels changed per drill not per session; weak first scores inflated retention; discarding a session bypassed the gate wait
Medium	9	8	Free-recall probes cut at 10 items; scaffolded drills inflated mastery; Major decoder misread words ending in g or c; speed ignored in timed gates; small content pools; quarterly re-test landed on gate days
Low	14	12	Contrast, focus, mobile overlap, delete race, false "Wren has grown" message

12.4 Unresolved after three rounds

Category	Why it is below 8	What would raise it
Curriculum	Early weeks repeat near-identical drills; sessions run about 16 to 28 minutes, not 30	More drill variety in Months 1 to 3; tune extra rounds to reach 30
Creative	Only Prologue and Chapters 1 to 3 are written in full; 9 chapters are one-line summaries	Write the remaining chapters and Calder ledger fragments
UI/UX	Faces can differ only by clothing; no 3D palace walk yet	Stronger face generator; optional 3D palace view

13. Prototype status, how to test, and publication plan

The prototype is a single 295 KB HTML file that runs offline in any modern browser. All 9 exercise templates and all 52 weeks run from the curriculum data. Nothing has been published or pushed anywhere.

13.1 What is implemented

Area	Status
Onboarding, local profile, alias and sigil, themes	Working
Day-1 baseline battery and quarterly repeats (days 91, 182, 273, 364)	Working
Full, recovery, review, assessment, gate and free-practice sessions	Working
9 templates: word recall, linked pairs, story chain, palace walk (forward and reverse), digits, Major conversion, names and faces, vocabulary, facts	Working
Interference, timed and guided variants	Working
Adaptive rules R1 to R10, error diagnosis, coaching cards	Working
Spaced delayed-recall probes at 1, 7, 30 (gates to 180) days	Working
MTS and five component scores, skill radar, trends, heatmap, baseline comparison	Working
Lumen, ranks, 14 achievements, world map, Wren's six-stage arc	Working
Palace builder, personal Major System list	Working
Export, import with validation, delete, cross-tab safety, error screen	Working
Story text	Prologue and Chapters 1 to 3 in full; Chapters 4 to 12 as summaries

Area	Status
Service worker and install	In the GitHub Pages build only
3D palace walk, cloud sync, notifications	Not built (later phases)

13.2 How to test

1. Download **mnemos-prototype.html** and open it in Chrome, Edge or Firefox on a phone or computer.
2. Create a profile and complete day 1 (about 20 minutes).
3. Close and reopen the file: progress should still be there.
4. To test next-day recall without waiting, go to **Keeper → Prototype review tools → Advance 1 day**, then start day 2.
5. Use **Jump curriculum to week** to try the palace (week 5), faces (week 9), numbers (week 14) and vocabulary (week 22) weeks.
6. Try **Export backup**, then **Delete all data**, then import the file from the start screen.
7. Leave comments directly on this document, or reply in chat.

Data saved while using the downloaded file stays in that browser on that device.

13.3 GitHub publication plan (on hold until you approve)

1. Create a public repository with the source: MIT licence for code, CC BY 4.0 for story and art.
2. Add a GitHub Actions workflow: install, run the 28 unit tests and the end-to-end test, build the static site.
3. Deploy the `dist` folder to GitHub Pages; the service worker makes it installable and offline-capable.
4. Run a trademark check on the name "Mnemos" and verify the content facts flagged in Section 11.
5. Add a privacy page (no data collected), a contribution guide and an issue template for curriculum feedback.

None of these steps will run without your explicit approval.

13.4 Version history

Version	Date	Changes
0.2	2026-11-01	Full story: prologue and 12 chapters in long-form prose; 52 weekly tales; 12 gate encounters; 24 side quests tied to real skill; 16 lore entries; 9 characters with staged reveals; Veyra tab rebuilt as Map, Quests, Chapters, People and Lore; all 52 weeks of lessons, drill instructions, onboarding and Wren's dialogue rewritten in plain, warmer language. 31 unit tests and the end-to-end test pass.
0.1	2026-11-01	First prototype; three independent audit rounds.

Version 0.3 (2026-11-01), pixel-art game edition. Built with 5 specialist agents:

- **Faces and portraits:** realistic generated pixel faces (10 skin tones, 21 hair styles, distinct features, matched to names) and 9 named character portraits with moods.
- **World art:** an animated pixel map, a title scene and 14 district banners.
- **Relics:** 76 collectibles across artifacts, ledger pages, trinkets, lost treasures and seals, each with its own icon.
- **The Undercroft:** an endless memory crypt. Keys are earned by real training, with lamp-oil lives, chests, 12 deep treasures and 15 lore tablets.
- **Game interface:** pixel fonts (Pixelify Sans, Silkscreen and VT323, all free for commercial use), dialogue boxes, a status bar, unlock pop-ups and generated retro game sound.

All 73 automated tests and the full end-to-end run pass.